23UXD2012 | Semester 4

Design For Special Needs

• Drashi Jain • Jagruti Ghuge • Krisha Karani

Introduction

Amputees are people who have gone through the amputation surgery and have lost a limb. There are many mental and emotional changes that they go through and it can affect basic mobility, independence, career prospects, and even self-identity.

There are many people living their lives after having their limbs amputed, our aim is not only to understand the physical, mental and emotional changes that these amputees go through but also get a wholistic view of the impact this has on the people around them and how they deal and process this happening in their lives.

Secondary research

Interview Insights

1. our main insight was that amputees had major concerns over **affordability, comfort** and budget of the prosthetics

2. our main insight was **multiple factors go into consideration** while selecting the prosthetic and prioritising the right factors can help prevent buying the wrong match of prosthetics

3. our main insight was **people's beliefs and support systems play a major role** in their emotional and mental journey after such incidents

4. our main insight was while **modern tools help the amputees be more independent** and manage their daily lives easily, it is more important to maintain good emotional support around oneself

5. our main insight was **access to a variety** of options of prosthetics and assistive devices can help make one's life easier

Problem Areas

Problem Area 1

After the amputation, choosing the right prosthetic is very important. With growing technology ands access there is an overwhelming number of options available to the patients, with no clear guidance from one reliable source on which to select. This leads to people making huge medical and personal decisions while still adjusting to life post-amputation, with little support and lots of pressure.

Along with the physical recovery, the emotional and psychological recovery is extremely important as well. Which often gets overlooked due to various reasons. Apart from the mandatory therapy patients don't have the comfort and safe space to speak freely about these aspects.

Problem Area 2

Ideation Mapping

Grouped Ideas – 4

Idea 1

The first idea was a **kiosk+website** digital product which focuses on the user (amputees) finding the right prosthetic for them by streamlining and grouping all the factors taken into consideration while selecting the same.

Idea 2

The second idea gives the user a **platform** where they can browse through all the prosthetics available in the market and filter according to what they want

Idea 3

The third idea suggests a website+app which will be the one stop destination for any mental health support that the user may need, with **community support** and guidance to get the right help

Idea 4

The fourth idea focused on the **interaction** between amputees, medical professionals and non-amputees in a casual way and focusing on mental health and being more open about the same.

Final Problem statement

After an amputation, patients **struggle to find** the right prosthetic suited to their **budget, lifestyle, and medical needs.** With **scattered, technical information** and little personalised guidance, they're often forced to make **overwhelming decisions** without clarity or confidence.

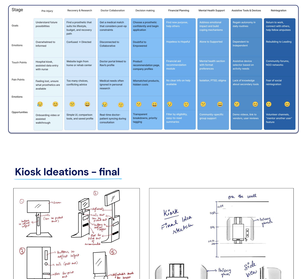
Final Concept

What system/platform are we proposing?
We're proposing Amputinity, a digital platform available primarily through an in-hospital kiosk and additionally as a website. The kiosk allows amputees to enter lifestyle, medical conditions and financial details while they wait, with staff assistance if needed. Doctors can access this and give input during consultations using the patient's unique ID. The web version acts as a secondary access for patients to revisit and update inputs from home. Amputinity offers tailored prosthetic recommendations, daily support tools, and mental health resources.

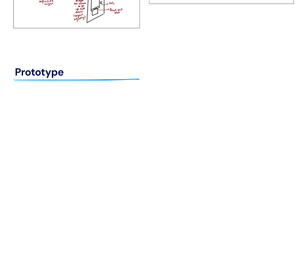
How does it work?
Patients receive a unique ID to log into the kiosk or website. They enter details like career, mobility requirements, hobbies, type of amputation and budget. Doctors input medical constraints. The system then generates a ranked list of prosthetics with tags like "sport-approved" or "budget-friendly," including details like pricing, delivery timeline and maintenance. It also suggests funding options and supportive tools, along with access to tools and mental health resources, aiming to holistically support amputees through recovery and reintegration.

How does it help the current problem at hand?
Amputinity simplifies a complex and overwhelming decision by combining personal, lifestyle, and medical inputs to provide clear, patient-specific prosthetic recommendations. It avoids the confusion of generic product ratings, supports decision-making with doctor-backed suggestions, and includes funding and mental health support. The platform, combining kiosk and website, bridges gaps for those with limited tech access, informational gaps and supports more confident, informed, and timely decisions.

User Journey Map



Kiosk Ideations – final



Prototype

THANK YOU